

Inequalities in One Variable

You can represent an inequality by using an inequality symbol, interval notation, or a graph.

Definitions

A **bounded interval** is the set of all real numbers between two numbers, called **endpoints**; the endpoints may or may not be included.

An **unbounded interval** is the set of all real numbers on one side of a number, called an endpoint; the endpoint may or may not be included.

A **bounded closed interval** includes both endpoints. This bounded closed interval is the set of all real numbers between -2 and 2 , including both -2 and 2 .



A **bounded open interval** does not include either endpoint. This bounded open interval is the set of all real numbers between -2 and 2 , excluding both -2 and 2 .



A **bounded half-open interval** includes one and only one endpoint. This bounded half-open interval is the set of all real numbers between -2 and 2 , including -2 but excluding 2 .



Graphing Interval Notation

A simple inequality using $<$, $>$, $\leq$, or $\geq$ indicates an unbounded interval.

In interval notation, a bracket means the endpoint is included, and a parenthesis means the endpoint is not included.

► **Think About It** In interval notation, you can never put a bracket next to an infinity symbol because infinity is not a number that can be included in an interval.

Interval notation for $x > 4$ is $x \in (4, \infty)$. The ∞ symbol is read “positive infinity,” and it means that the interval is unbounded to the right.



Interval notation for $x \leq 1$ is $x \in (-\infty, 1]$. The $-\infty$ symbol is read “negative infinity,” and it means that the interval is unbounded to the left.



The table shows the four cases of unbounded intervals.

Inequality notation	Graph	Interval notation
$x > a$		$x \in (a, \infty)$
$x \geq a$		$x \in [a, \infty)$
$x < a$		$x \in (-\infty, a)$
$x \leq a$		$x \in (-\infty, a]$

In interval notation, the set of all real numbers is $(-\infty, \infty)$.

► **Remember** The symbol $\in$ is read “is an element of.”

► **Think About It** In interval notation, always put the lesser number on the left.

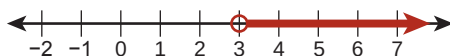
EXAMPLE 1

Write the inequality in interval notation, and then graph.

A $x > 3$

SOLUTION

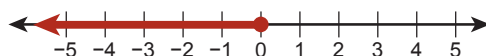
Interval notation for $x > 3$ is $x \in (3, \infty)$. The interval is unbounded to the right, and does not include the left endpoint.



B $m \leq 0$

SOLUTION

Interval notation for $m \leq 0$ is $m \in (-\infty, 0]$. The interval is unbounded to the left, and includes the right endpoint.



C $t > -2.5$

SOLUTION

Interval notation for $t > -2.5$ is $t \in (-2.5, \infty)$. The interval is unbounded to the right, and does not include the left endpoint.

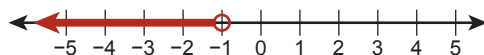


Writing Interval Notation for a Given Graph

EXAMPLE 2

Write an inequality in interval notation for the graph.

A

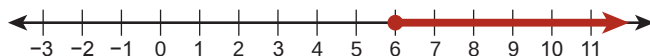


SOLUTION

The graph is unbounded to the left. It is a ray pointing toward “negative infinity.” The open circle indicates that -1 is not included in the set.

$$x \in (-\infty, -1)$$

B



SOLUTION

The graph is unbounded to the right. It is a ray pointing toward “positive infinity.” The closed circle indicates that 6 is included in the set.

$$x \in [6, \infty)$$

► **Think About It** The symbols ∞ and $-\infty$ do not denote real numbers. They simply denote unbounded intervals.

Solving an Inequality in One Variable with a Restricted Domain

When you solve inequalities, the domain of the variable may be whole numbers, integers, or real numbers.

EXAMPLE 3

Solve the inequality over the domain of whole numbers.
Graph the solution set.

A $\frac{x}{2} - 1 \leq 1$

SOLUTION

Solve an inequality as you would solve an equation, but reverse the inequality sign when you multiply or divide by a negative number.

$$\frac{x}{2} - 1 \leq 1 \quad \text{Original inequality}$$

$$\frac{x}{2} \leq 2 \quad \text{Add 1 to both sides.}$$

$$x \leq 4 \quad \text{Multiply both sides by 2.}$$

$$x \in \{0, 1, 2, 3, 4\}$$



B $-2 < 4 - a$

SOLUTION

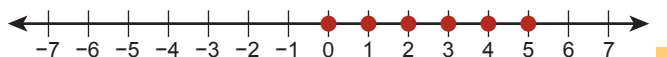
$$-2 < 4 - a \quad \text{Original inequality}$$

$$-6 < -a \quad \text{Subtract 4 from both sides.}$$

$$\frac{-6}{-1} > \frac{-1 \cdot a}{-1} \quad \text{Divide both sides by } -1, \text{ reversing the inequality sign.}$$

$$6 > a$$

$$a \in \{0, 1, 2, 3, 4, 5\}$$



EXAMPLE 4

Solve the inequality over the domain of integers. Graph the solution set.

A $\frac{n}{3} + 3 > 4$

SOLUTION

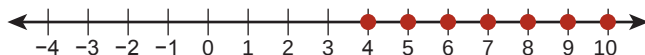
$$\frac{n}{3} + 3 > 4 \quad \text{Original inequality}$$

$$\frac{n}{3} > 1 \quad \text{Subtract 3 from both sides.}$$

$$n > 3 \quad \text{Multiply both sides by 3.}$$

$$n \in \{4, 5, 6, \dots\}$$

The graph is the set of points with coordinates that are integers greater than 3.



B $-7x \geq 3 - 2x$

SOLUTION

$$-7x \geq 3 - 2x \quad \text{Original inequality}$$

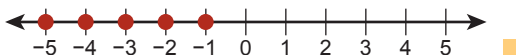
$$-5x \geq 3 \quad \text{Add } 2x \text{ to both sides.}$$

$$\frac{-5x}{-5} \leq \frac{3}{-5} \quad \text{Divide both sides by } -5, \text{ reversing the inequality sign.}$$

$$x \leq -\frac{3}{5}$$

$$x \in \{\dots, -3, -2, -1\}$$

The graph is the set of points with coordinates that are integers less than or equal to $-\frac{3}{5}$.



EXAMPLE 5

Solve the inequality over the domain of integers. Graph the solution set.

A $x + 6 \geq 3(x + 1) - 11$

SOLUTION

$$x + 6 \geq 3(x + 1) - 11 \quad \text{Original inequality}$$

$$x + 6 \geq 3x + 3 - 11 \quad \text{Distribute 3.}$$

$$x + 6 \geq 3x - 8$$

$$x + 14 \geq 3x \quad \text{Add 8 to both sides.}$$

$$14 \geq 2x \quad \text{Subtract } x \text{ from both sides.}$$

$$7 \geq x \quad \text{Divide both sides by 2.}$$

$$x \in \{\dots, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$$

The graph is the set of points with coordinates that are integers less than or equal to 7.



► **Remember** $7 \geq x$ is the same as $x \leq 7$.

B $-9 - 3x > 3 - 5x$

SOLUTION

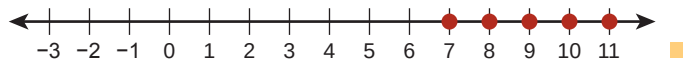
$$-9 - 3x > 3 - 5x \quad \text{Original inequality}$$

$$-3x > 12 - 5x \quad \text{Add 9 to both sides.}$$

$$2x > 12 \quad \text{Add } 5x \text{ to both sides.}$$

$$x > 6 \quad \text{Divide both sides by 2.}$$

$$x \in \{7, 8, 9, 10, 11, \dots\}$$



Real-World Inequalities

You can describe many real-world situations using inequalities. First, identify the unknown quantity. Next, write an inequality to model the situation and then solve.

EXAMPLE 6

Marcus is planning a deep-sea fishing trip. For boat rental, there is an initial fee of \$150 plus \$70 for each hour the boat is used. How many hours can Marcus rent a boat if he has \$395?

SOLUTION

Marcus will be charged \$70 times the number of hours, x , plus \$150. This sum must be less than or equal to the amount he has, \$395.

Write an inequality to describe the situation. Then solve for x .

$$70x + 150 \leq 395 \quad \text{Use } x \text{ to represent the number of hours.}$$

$$70x \leq 245 \quad \text{Subtract 150 from both sides.}$$

$$x \leq 3.5 \quad \text{Divide both sides by 70.}$$

Marcus can rent the boat for 3.5 h or less. ■

► **Think About It** If Marcus rents a boat for exactly 3.5 h, he will be charged $70(3.5) + 150 = 395$, or \$395, which is the exact amount he has. He could rent the boat for fewer than 3.5 h and have money left over.